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BACHELOR OF COMPUTER APPLICATION

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LECTURE NOTE

Research Methodology (BCAC702)

Module 1: Foundations of Research

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1 Introduction and Scope of Module 1

Research, at its most fundamental level, is the disciplined pursuit of knowledge. It is not merely “looking something up” or “reading a few papers.” It is a *systematic, controlled, empirical*, and *critical* investigation of a phenomenon with the express purpose of discovering new facts, verifying existing ones, or building theoretical frameworks that explain how the world works [1].

For students of Computer Applications, the relevance of research methodology cannot be overstated. Whether one is designing a new intrusion-detection algorithm, evaluating the usability of a software interface, or measuring the performance gap between two machine-learning classifiers, the *logic of inquiry* remains the same. Module 1 lays the philosophical and procedural groundwork upon which every subsequent module—and indeed every research project a student will ever undertake—is built.

Key Point: What this module covers

1. Meaning, objectives, and types of research
2. The research process and characteristics of good research
3. Identifying and defining a research problem
4. Formulation of research questions and objectives
5. Variables in research and research motivation

2 Meaning of Research

2.1 Etymological and Formal Definitions

The word *research* derives from the Middle French *rechercher*, meaning “to search again” or “to investigate thoroughly.” The prefix *re-* signals repetition and rigour: a researcher does not glance at a problem once and declare a conclusion; she returns to the evidence again and again until the picture is clear [2].

Several authoritative definitions deserve attention:

Definition – C. R. Kothari [1]

“Research in common parlance refers to a search for knowledge. ... In a broader sense, research is a movement from the known to the unknown. ... Research is an academic activity and, as such, the term should be used in a technical sense.”

Definition – Uma Sekaran & Roger Bougie [3]

“Research is an organised, systematic, data-based, critical, objective, scientific inquiry or investigation into a specific problem, undertaken with the purpose of finding answers or solutions to it.”

Definition – J. W. Creswell [4]

“Research is a process of steps used to collect and analyse information to increase our understanding of a topic or issue. ... It constitutes a study of a problem using rigorous, systematic procedures.”

Three ideas recur across these definitions and are worth isolating:

- (i) **Systematic procedure.** Research follows a defined sequence of steps; it is not haphazard.
- (ii) **Empirical grounding.** Conclusions rest on observable evidence, not on authority or intuition alone.
- (iii) **Critical scrutiny.** Findings are subjected to peer review, replication, and falsification attempts.

2.2 Research versus Everyday Inquiry

A natural question arises: how does formal research differ from the curiosity we all exercise daily? Table 1 summarises the distinction.

Table 1: Formal Research vs. Everyday Inquiry

Dimension	Everyday Inquiry	Formal Research
Purpose	Satisfy immediate curiosity	Generate generalisable knowledge
Method	Ad hoc, informal	Systematic, replicable protocol
Data	Anecdotal, selective	Collected under controlled conditions
Analysis	Subjective interpretation	Statistical / logical rigour
Reporting	Casual conversation	Peer-reviewed publication
Bias control	Rarely addressed	Explicit safeguards (blinding, randomisation)

Example: Everyday vs. Research Inquiry in Computing

Everyday: “My laptop feels slower after the last Windows update.”

Research: “A controlled experiment measuring mean boot-time and application-launch latency ($n = 50$ machines) before and after the Windows 11 23H2 update, analysed with a paired-sample t -test at $\alpha = 0.05$.”

3 Objectives of Research

Why do we conduct research at all? Kothari [1] identifies four broad objectives, which we expand below.

1. **Exploration and Discovery.** To gain familiarity with a phenomenon or to achieve new insights into it. In computing, this might mean investigating whether quantum annealing can outperform classical simulated annealing on a specific class of NP-hard scheduling problems.
2. **Description.** To portray accurately the characteristics of a particular individual, situation, or group. A descriptive study might catalogue the frequency and types of phishing attacks reported by Indian SMEs during 2024–2025.
3. **Verification and Hypothesis Testing.** To determine the frequency with which something occurs or its association with something else. For instance, testing whether the adoption of multi-factor authentication is statistically associated with a reduction in account-compromise incidents.
4. **Theory Building and Explanation.** To test a causal relationship between variables. An example would be developing a theoretical model that explains how developer experience (DX) influences software defect density in agile teams.

Key Point: Hierarchy of Objectives

The four objectives form a natural progression: *Exploration* → *Description* → *Verification* → *Explanation*. Early-stage research in a new domain tends to be exploratory; mature fields move toward causal explanation.

4 Types of Research

Research can be classified along several axes. The syllabus emphasises four principal types: *Basic (Fundamental)*, *Applied*, *Exploratory*, and *Descriptive*. In practice, these categories overlap, and a single study may serve more than one purpose [2]; Figure 1 shows how the four types relate.

4.1 Basic (Fundamental / Pure) Research

Basic research is driven by intellectual curiosity rather than an immediate practical need. Its goal is to expand the frontiers of knowledge. The researcher asks, “*What is there to know?*” rather than “*How can I fix this?*”

Example: Basic Research in CS

Alan Turing’s 1936 paper “On Computable Numbers” was pure research. Turing was not building a product; he was asking whether every mathematical problem could, in principle, be solved by a mechanical procedure. The theoretical framework he created became the foundation of all modern computing [5].

4.2 Applied Research

Applied research seeks solutions to specific, practical problems facing an individual, organisation, or society. It draws on the theories developed by basic research and tests them in real-world settings [3].

Example: Applied Research in CS

Developing and evaluating a convolutional-neural-network model for early detection of diabetic retinopathy from fundus photographs in rural Indian clinics is applied research: it targets a concrete healthcare problem and measures success by clinical utility.

4.3 Exploratory Research

Exploratory (or formulative) research is undertaken when the problem is not clearly defined. Its purpose is to gain familiarity, generate hypotheses, and identify variables for later study. Methods include literature surveys, expert interviews, and pilot case studies [1].

4.4 Descriptive Research

Descriptive research includes surveys and fact-finding enquiries of different kinds. The major purpose is description of the state of affairs as it exists at present. The researcher has no control over the variables; she can only report what has happened or what is happening [4].

4.5 Comparative Summary

Table 2: Comparison of Research Types

Type	Primary Goal	Typical Question	CS Example
Basic	Expand knowledge	“What is com-putable?”	Complexity theory, formal languages
Applied	Solve a problem	“How to reduce la-tency?”	Optimising CDN caching strategies
Exploratory	Generate hypothe-ses	“What factors mat-ter?”	Pilot study on AI ethics perceptions
Descriptive	Portray status quo	“What is happen-ing?”	Survey of cloud-adoption rates in Indian startups

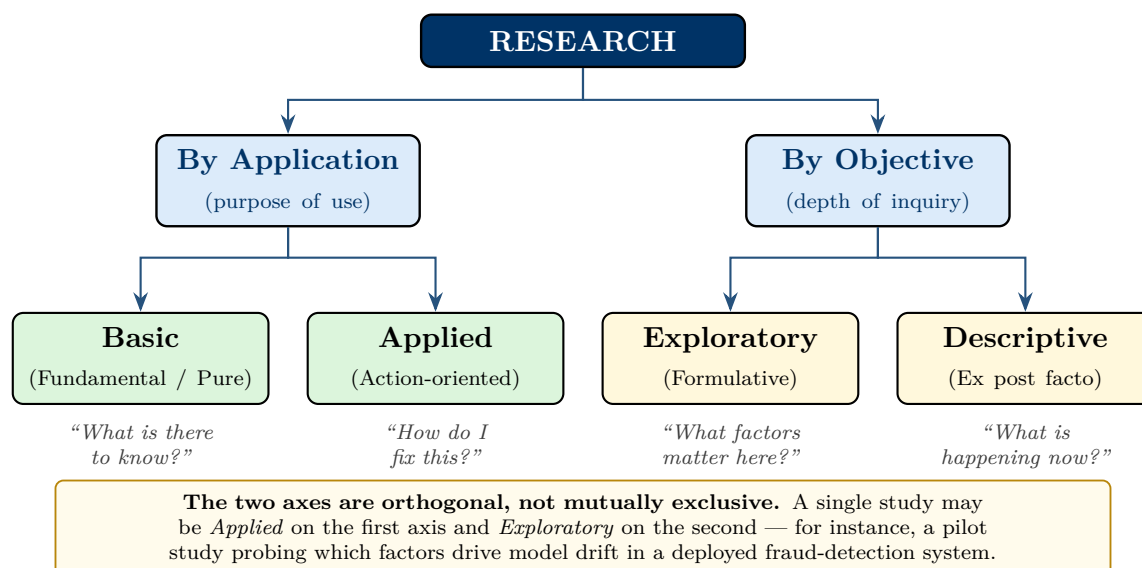


Figure 1: The four research types emphasised in the syllabus, arranged along two *independent* classification axes. Basic–Applied separates studies by *why* they are undertaken; Exploratory–Descriptive separates them by *how far* the inquiry has progressed. Because the axes are independent, the boundaries are porous [1, 2].

5 The Research Process

Research is not a single act of inspiration; it is a *process*—a sequence of interdependent steps that, when followed carefully, lend credibility and replicability to the findings. Kothari [1] and Kumar [2] both stress that the steps are iterative rather than strictly linear: one often loops back to refine earlier decisions in light of later discoveries.

5.1 The Standard Steps

- Step 1: Formulating the Research Problem.** Identifying a gap, a contradiction, or a practical need and stating it precisely (see Section 7).
- Step 2: Extensive Literature Review.** Surveying journals, conference proceedings, books, and credible online sources to understand what is already known and to avoid duplication.
- Step 3: Developing Objectives and Hypotheses.** Translating the problem into measurable objectives and, where appropriate, testable hypotheses.
- Step 4: Preparing the Research Design.** Deciding on the overall strategy: experimental, survey, case-study, simulation, etc.
- Step 5: Determining Sampling Design.** Selecting the population, sample size, and sampling technique (probability or non-probability).
- Step 6: Collecting Data.** Administering questionnaires, conducting experiments, scraping datasets, or running simulations.
- Step 7: Executing the Project / Data Collection.** Carrying out the plan while maintaining quality control and ethical standards.

Step 8: Analysis of Data. Coding, tabulating, and applying statistical or computational techniques to extract patterns.

Step 9: Hypothesis Testing and Interpretation. Using inferential statistics (t -tests, χ^2 , ANOVA, etc.) to accept or reject hypotheses and to interpret results in context.

Step 10: Reporting and Presentation. Writing the thesis, paper, or report following a standard format (IEEE, APA, etc.) and defending the work before peers.

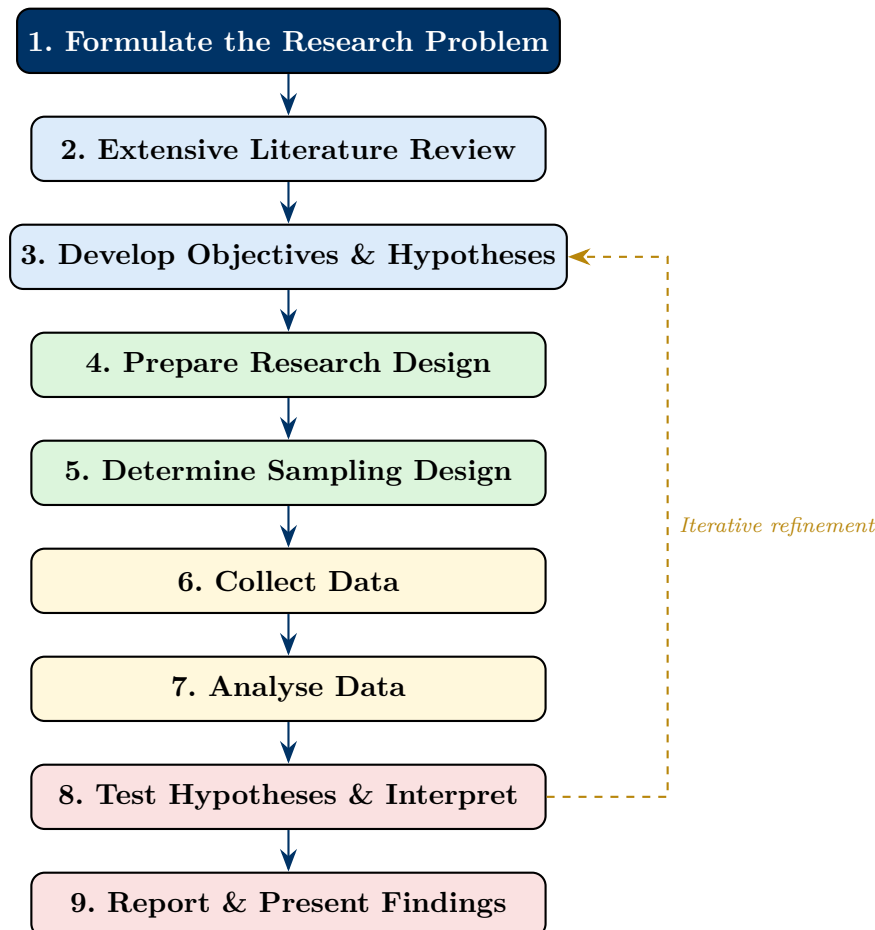


Figure 2: The research process as a sequence of iterative steps. The dashed gold arrow emphasises that hypothesis testing often forces the researcher to revisit objectives or design [1].

6 Characteristics of Good Research

Not every investigation qualifies as *good* research. Kothari [1] and Creswell [4] enumerate several hallmarks. We summarise them below and then illustrate with a computing example.

- (a) **Systematic.** Follows a structured, logical sequence.
- (b) **Controlled.** Extraneous variables are held constant or accounted for, so that observed effects can be attributed to the variable under study.

- (c) **Empirical.** Conclusions are grounded in observable data, not in authority or speculation.
- (d) **Objective.** The researcher minimises personal bias through blinding, randomisation, and peer review.
- (e) **Replicable.** Another researcher, following the same protocol, should be able to reproduce the results.
- (f) **Valid and Reliable.** The instruments measure what they claim to measure (validity) and yield consistent results across repetitions (reliability).
- (g) **Critical.** The process and findings withstand scrutiny; potential errors are acknowledged and addressed.
- (h) **Ethical.** Informed consent, data privacy, and intellectual honesty are maintained throughout.

Example: Good vs. Poor Research Practice

Poor: A student benchmarks two sorting algorithms on a single input of size $n = 100$, runs each once, and concludes that Algorithm A is “always faster.”

Good: The student tests both algorithms on randomised inputs of sizes $n \in \{10^3, 10^4, 10^5, 10^6\}$, repeats each trial 30 times, reports mean and standard deviation of execution time, controls for hardware and OS, and applies a Mann–Whitney U test before drawing conclusions.

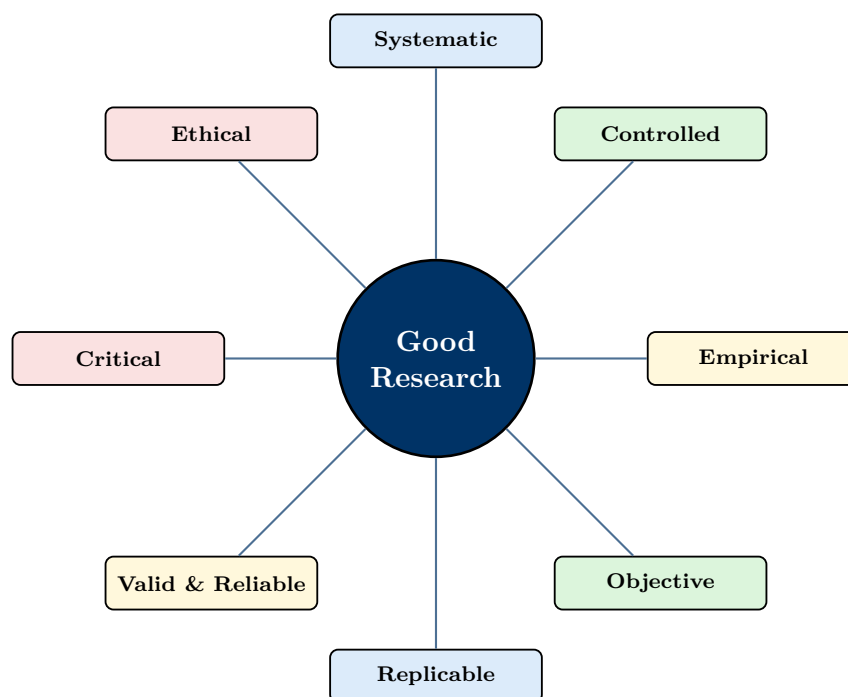


Figure 3: Eight hallmarks of good research radiating from the central concept [1, 4].

7 Identifying and Defining a Research Problem

7.1 Why the Problem Comes First

Albert Einstein reportedly remarked that if he had one hour to save the world, he would spend fifty-five minutes defining the problem and five minutes on the solution. In research methodology, the problem statement is the anchor: every subsequent decision—design, sampling, analysis, interpretation—flows from it [2].

7.2 Sources of Research Problems

- **Personal experience and observation.** A developer notices that code-review turnaround time spikes after sprint boundaries.
- **Literature gaps.** A survey paper notes that “no study has examined the effect of X on Y in the Indian context.”
- **Practical / industry needs.** A hospital requests a predictive model for patient readmission.
- **Theoretical contradictions.** Two published models yield conflicting predictions about network throughput under load.
- **Government and funding priorities.** National missions (e.g., Digital India, AI for All) highlight areas needing investigation.

7.3 Criteria for Selecting a Good Problem

Sekaran and Bougie [3] suggest the following filters:

Table 3: Criteria for a Well-Chosen Research Problem

Criterion	Explanation
Relevance	Addresses a genuine gap or practical need.
Feasibility	Can be investigated within available time, budget, and expertise.
Specificity	Narrow enough to be tractable; not “study all of AI.”
Novelty	Adds something new; does not merely replicate without justification.
Ethical viability	Can be pursued without harming participants or violating norms.
Measurability	Key variables can be operationalised and measured.

7.4 From a Vague Interest to a Precise Problem Statement

Example: Refining a Research Problem

Vague interest: “Cyber security is important.”

Narrower: “Phishing attacks affect Indian banking customers.”

Research problem: “What is the effect of a gamified security-awareness training programme on the click-through rate of simulated phishing emails among retail-banking customers aged 18–35 in Kolkata, measured over a 12-week period?”

Notice how the final statement specifies the *who*, *what*, *where*, *when*, and *how* of the investigation.

8 Formulation of Research Questions and Objectives

8.1 Research Questions

A research question (RQ) translates the problem into an interrogative form that the study will answer. Good RQs are:

- **Focused**—addressing a single, well-defined issue.
- **Researchable**—answerable through data collection and analysis, not through moral judgement.
- **Complex enough**—requiring investigation rather than a yes/no answer.
- **Aligned**—directly traceable to the problem statement [4].

Example: Research Questions

- **RQ1:** Does the gamified training programme reduce the phishing click-through rate compared to a conventional text-based module?
- **RQ2:** Is the effect moderated by the participant’s prior experience with online banking?
- **RQ3:** How do participants perceive the usability of the gamified intervention?

8.2 Research Objectives

Objectives convert questions into actionable, measurable targets. The **SMART** framework is widely recommended [2] and is set out in Figure 4:

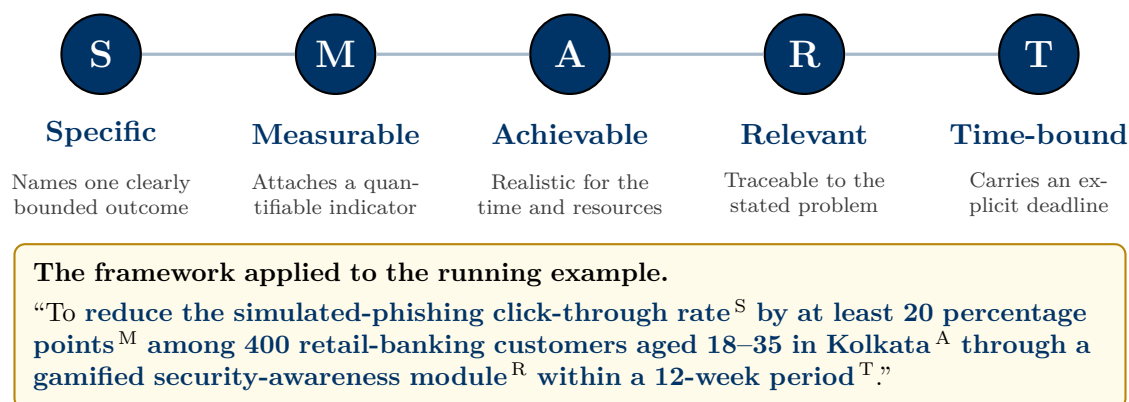


Figure 4: The SMART framework for converting a research question into a measurable objective. Each criterion is mapped onto the running phishing-awareness example; strip any one of the five clauses and the statement decays from an objective back into an aim [2].

Note: Common Pitfall

Students often confuse *aims* (broad aspirations) with *objectives* (specific, measurable targets). “To improve cyber security” is an aim; “To reduce the phishing click-through rate by at least 20% within 12 weeks” is an objective.

9 Variables in Research

A *variable* is any characteristic that can take on different values. Identifying and classifying variables is essential because the entire research design—from data collection to statistical analysis—revolves around them [3]. Figure 5 sets out how the six types interlock.

9.1 Principal Types of Variables

Independent Variable (IV).

The presumed *cause*; the variable the researcher manipulates or categorises. In an experiment comparing two training methods, the *type of training* is the IV.

Dependent Variable (DV).

The presumed *effect*; the outcome measured. Here, the *phishing click-through rate* is the DV.

Control Variable.

A variable held constant to prevent confounding. For instance, keeping the *duration of training* identical across groups.

Intervening (Mediating) Variable.

A variable that explains *how* or *why* the IV affects the DV. For example, *security self-efficacy* may mediate the relationship between training and click-through rate.

Moderating Variable.

A variable that affects the *strength* or *direction* of the IV–DV relationship. *Prior banking experience* might moderate the training effect.

Extraneous Variable.

Any uncontrolled variable that could influence the DV and threaten internal validity.

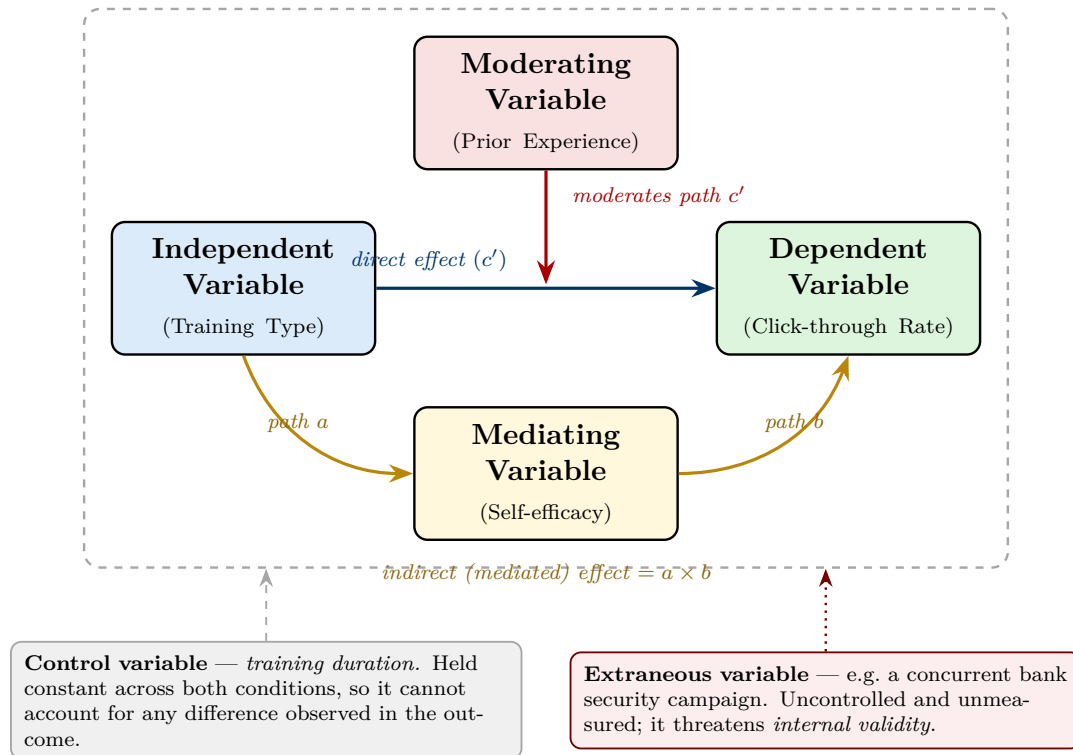


Figure 5: Schematic of variable relationships in the hypothetical phishing-awareness study. **Blue** marks the direct effect c' ; **gold** marks the indirect route through the mediator, whose magnitude is the product $a \times b$; **red** marks moderation, which acts on the *path* rather than on a variable, and so is drawn terminating on the arrow itself. The dashed envelope is the study design: the control variable is held constant inside it, while extraneous variables intrude on it from outside. The total effect therefore decomposes as $c = c' + ab$ [3].

9.2 Operationalisation

Naming a variable is not enough; the researcher must define *how* it will be measured. This process is called *operationalisation* [4].

Table 4: Operationalisation Example

Variable	Conceptual tion	Defini- tion	Operational Definition
Training type	The pedagogical approach used	ap-	0 = text-based module; 1 = gamified module
Click-through rate	Proportion of phishing emails on which the user clicks		$\frac{\text{clicks}}{\text{emails delivered}} \times 100$ over 12 weeks
Prior experience	Familiarity with online banking		Likert-scale score (1–5) on a 6-item questionnaire

10 Research Motivation

10.1 What Drives Research?

Research is demanding, often tedious, and frequently inconclusive. Understanding *why* researchers persist is itself a topic of scholarly interest. Kothari [1] and Singh [5] identify both intrinsic and extrinsic motivators.

(a) Intrinsic Motivation

- Intellectual curiosity and the joy of discovery.
- Desire to contribute to the body of knowledge.
- Personal satisfaction in solving a difficult problem.

(b) Extrinsic Motivation

- Academic requirements (dissertation, thesis, degree).
- Career advancement, publications, and recognition.
- Funding, grants, and institutional mandates.
- Societal impact—solving problems that affect communities.

10.2 Motivation in the Context of BCA Students

For a BCA student, research motivation often begins with the project requirement. However, the most productive students transcend the extrinsic “I need to submit a project” mindset and develop genuine curiosity.

Key Point: Sustaining Motivation

- Choose a problem you care about; passion sustains effort through setbacks.
- Break the project into small, achievable milestones.
- Seek regular feedback from supervisors and peers.
- Read widely—attending seminars and reading groups exposes you to new questions.
- Accept that negative results are still results; they advance knowledge by eliminating false paths.

11 Module Summary

Module 1 at a Glance

- Research is a **systematic, empirical, and critical** inquiry aimed at generating or verifying knowledge.
- Its objectives span **exploration, description, verification, and theory building**.
- Principal types include **basic, applied, exploratory, and descriptive** research; the boundaries are fluid.
- The **research process** comprises ten iterative steps, from problem formulation to reporting.
- Good research is **systematic, controlled, empirical, objective, replicable, valid, reliable, critical, and ethical**.
- A well-defined **problem statement** specifies the who, what, where, when, and how of the investigation.
- **Research questions** and **SMART objectives** translate the problem into actionable targets.
- **Variables** (independent, dependent, mediating, moderating, control, extraneous) must be identified and **operationalised**.
- **Motivation**—both intrinsic and extrinsic—sustains the researcher through the rigours of the process.

Review Questions

- Q1.** Define research in your own words. How does it differ from casual, everyday inquiry? Illustrate with a computing example.
- Q2.** Distinguish between basic and applied research. Can a single study be both? Justify with an example from machine learning.
- Q3.** List and explain any five characteristics of good research.
- Q4.** Describe the ten steps of the research process. Why is the process described as iterative rather than strictly linear?
- Q5.** Identify a research problem relevant to cyber security. Refine it from a vague interest into a precise problem statement.
- Q6.** Formulate two research questions and three SMART objectives for the problem stated in Q5.
- Q7.** For the study in Q5, identify the independent, dependent, mediating, moderating, and control variables. Draw a schematic diagram showing their relationships.

Q8. Discuss the difference between intrinsic and extrinsic research motivation. Which do you consider more sustainable for a long-term research career, and why?

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